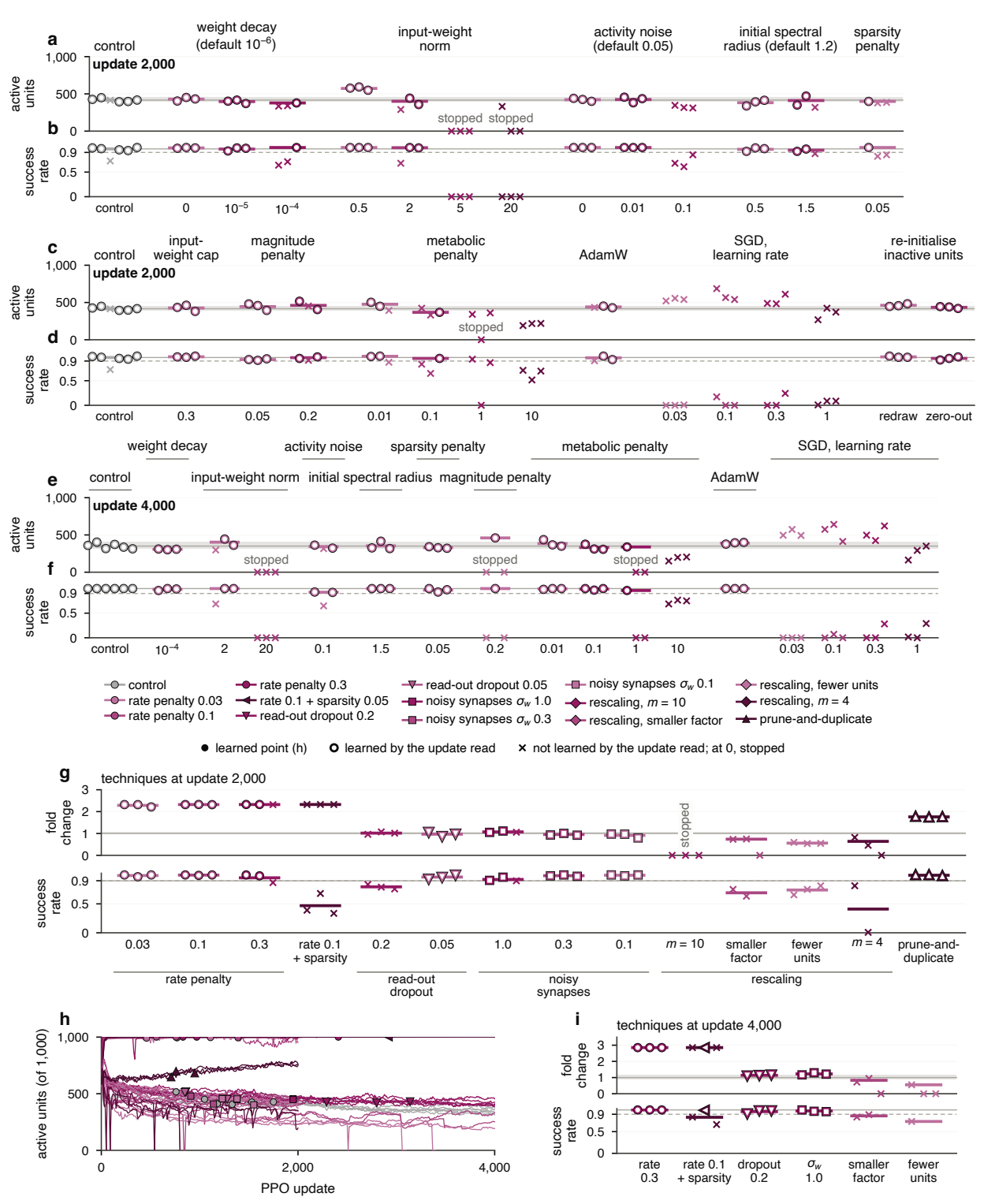

a
control
weight decay (default 10^{-6})
input-weight norm
activity noise (default 0.05)
initial spectral radius (default 1.2)
sparsity penalty
update 2,000
active units
1,000
500
0
b
success rate
0.9
0.5
0
stopped
control 0 10^{-5} 10^{-4} 0.5 2 5 20 0 0.01 0.1 0.5 1.5 0.05
c
control
input-weight cap
magnitude penalty
metabolic penalty
AdamW
SGD, learning rate
re-initialise inactive units
update 2,000
d
stopped
control 0.3 0.05 0.2 0.01 0.1 1 10 0.03 0.1 0.3 1 redraw zero-out
weight decay
activity noise
sparsity penalty
metabolic penalty
SGD, learning rate
e
control
input-weight norm
initial spectral radius
magnitude penalty
AdamW
update 4,000
f
stopped
control 10^{-4} 2 20 0.1 1.5 0.05 0.2 0.01 0.1 1 10 0.03 0.1 0.3 1
control
rate penalty 0.03
rate penalty 0.1
rate penalty 0.3
rate 0.1 + sparsity 0.05
read-out dropout 0.2
read-out dropout 0.05
noisy synapses σ_w 1.0
noisy synapses σ_w 0.3
noisy synapses σ_w 0.1
rescaling, m = 10
rescaling, smaller factor
rescaling, fewer units
rescaling, m = 4
prune-and-duplicate
learned point (h)
learned by the update read
not learned by the update read; at 0, stopped
g
techniques at update 2,000
fold change
3
2
1
0
success rate
0.9
0.5
0
stopped
0.03 0.1 0.3 rate 0.1 + sparsity 0.2 0.05 1.0 0.3 0.1 m = 10 smaller factor fewer units m = 4 prune-and-duplicate
rate penalty
read-out dropout
noisy synapses
rescaling
h
active units (of 1,000)
1,000
500
0
0
2,000
4,000
PPO update
i
techniques at update 4,000
fold change
3
2
1
0
success rate
0.9
0.5
0
rate 0.3
rate 0.1 + sparsity
dropout 0.2
σ_w 1.0
smaller factor
fewer units